GNBF5020

Practice Examination 02 · Answer Key

Cumulative · Lectures 1–7 · answers, model answers and marking scheme

GNBF5020 · Introduction to Molecular Biology and Genetics

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Keys the paper: GNBF5020_Practice_Exam_02_Cumulative_L1-L7.pdf · 30 marks · 90 minutes

Built strictly from the course materials in this folder: the Lectures 1–5 revision pack, the Lecture 6 and Lecture 7 guides, the case study bank and the exam focus prediction

Part	Question type	Marks per question	Total
A	Multiple choice — the letter, plus why the distractors fail	2	10
B	Pipeline mapping — one mark per limb, as marked	6	12
C	Exception handling — 3 to find the break, 5 to follow it	8	8
	Total		**30**

[DO NOT OPEN FIRST] Do not open this before you have sat the paper. The question paper is a separate file — `GNBF5020_Practice_Exam_02_Cumulative_L1–L7.pdf`. Sit it closed-book first, then come back here. Reading the key before you attempt it removes exactly the thing the paper is for: finding out which parts of the wiring you actually have.

[HOW TO MARK] How to mark yourself. Every MCQ carries 2 marks: 1 for the letter, 1 for being able to say why it is right **and** why the wrong options fail. A right letter with no reasoning is half marks — because the reasoning is the part the exam actually tests. The pipeline and exception questions are marked limb by limb, and the limb marks are given in the model answers.

[AND AFTER YOU SUBMIT] This key is not the post-mortem. It gives the answers, the model answers and the marks. The full post-mortem — your errors, one by one, each biological mistake translated into computational or data-flow terms — is written when you submit your answers, so that it can address what *you* got wrong rather than what a stranger might.

How to use this key

Marks	What the mark is for
MCQ (2 each)	1 mark for the correct letter, 1 mark for being able to say **why** it is correct and **why each distractor fails**. A right letter with no reason is 1 mark, not 2.
Pipeline (6 each)	1 mark per limb, as marked below. Partial marks are earned by naming the right stage even if the reasoning around it is thin.
Exception (8)	3 marks for locating the break and tracing the consequence, 1 each for the four shorter limbs.

Total: 30 marks.

Everything below is drawn from the same course materials as the paper: the Lectures 1–5 revision pack (audio + slides), the Lecture 6 guide, the Lecture 7 exam-focus guide with its Appendix C corrections, the case study bank and the exam focus prediction.

PART A — Multiple choice: the key and the reasoning

The correct letter for each question, why it is correct, and — for every option that is wrong — the specific reason it fails. A right letter with no reason is half marks.

Q1 — Answer: **A**

Why A is right. The lecture splits this pair deliberately. The **biomarker** is the biological entity — here, HER2 overexpression on the tumour cell surface. The **companion diagnostic** is the *assay* that measures that entity in a clinic, and either platform can be the CDx, but only after it has been validated against the material the trial will use; choosing a platform does not change what the biomarker is. The CDx is not optional: it is developed **alongside** the drug, because the recruitment design screens incoming patients with it, so it has to be ready before recruitment opens. A biomarker without a validated CDx is a biological observation that nobody can act on; a CDx without a validated biomarker is an instrument measuring nothing meaningful.

Option by option — why each one stands or falls.

A.	is right — the biomarker is the biology, the CDx is the assay that measures it, and the CDx is developed alongside the drug and must be ready before recruitment. Full reasoning above.
B.	fails because it merges the two things the lecture separates. The biomarker is HER2 overexpression; IHC is the instrument that reads it. Choosing a platform does not settle the question, it moves it: which assay, validated how, against which tumour material, and does it classify the same patients the label will name?
C.	fails because a trial's evidence base is defined by how its patients were selected. A CDx filed after launch cannot be inserted retrospectively into a population selected by a different assay, or by none. The biomarker indication on a label has to name the assay, the cut-off and the validation that generated the data.
D.	fails because prognostic is not predictive. HER2 status says how a patient may do; it does not say that HER2-positive patients gain from this antibody while HER2-negative patients do not. Withholding a treatment needs a treatment arm and a biomarker-by-treatment interaction, not a survival curve.
E.	fails for two separate reasons. Convenience is not validation — a plasma assay has to be validated against tumour material and shown to classify the same patients before it can define the trial population. And the two readouts are different jobs: the selection CDx decides who enters the trial, while the resistance readout (EGFR T790M in plasma ctDNA) guides the next line after progression. A read-out validated for one job does not validate the other, and swapping the assay changes the population and therefore the evidence base for the label.

In systems terms. The biomarker is the data; the CDx is the reader that turns the data into a decision. The chain runs biology → assay → enrolment → outcome, and the assay sits in the middle: if it is missing, unvalidated or late, everything upstream (the biology, the drug) and everything downstream (the label, the treated population) is untouched and nothing

gets through. "The CDx is optional" is the classic mistake of assuming that the input and the output are adjacent — in a pipeline they are separated by the instrument that actually does the work.

Q2 — Answer: **C**

Why C is right. Two different questions are being answered by the same score, and each needs its own proof.

- **Prognostic.** The RS is an independent predictor of distant recurrence: **HR 2.81 (95 % CI 1.70–4.64), P < 0.001** in the derivation analysis (Paik et al., *N Engl J Med* 2004) — and note the scale, because it is the classic misreading: the hazard ratio is **per 50-point increment** of RS, not per point. Age, tumour size, grade and receptor status were not significant in that model, which is what makes the score usable on its own. The derivation and validation history matters too: the algorithm was **derived** in cohorts that included the tamoxifen arm of **NSABP B-20** (with two other cohorts, 447 patients) and was then **prospectively validated on archival NSABP B-14 blocks**, in which RT-PCR succeeded in **668 of 675** blocks. In that prospective set the 10-year distant recurrence was **6.8 % (4.0–9.6)** in the low band (**< 18**), **14.3 % (8.3–20.3)** in the intermediate band (**18 to < 31**) and **30.5 % (23.6–37.4)** in the high band (**31 to 100**); the score runs **0 to 100**.
- **Predictive.** This needs a treatment arm, because the question is differential — "does this patient gain from this drug?" is a comparison, and a comparison needs two arms. In **NSABP B-20** (Paik et al., *J Clin Oncol* 2006;24:3726–3734) **651** node-negative, ER+ patients were randomised to **tamoxifen alone (227)** or **tamoxifen plus chemotherapy (424)**; the biomarker-by-treatment interaction gave **P = 0.038**, the benefit of chemotherapy was confined to the **RS ≥ 31** group, and in the low-RS group the odds ratio for chemotherapy exceeded 1 — which is why low RS is not "harmless", it may be net harmful.

The panel itself is **21 genes by RT-qPCR** — **16** cancer-related plus **5** reference genes that normalise the assay — and the invasion group contains **MMP11** (whose protein alias is **ST3**) and **CTSL2**. The comparator is **MammaPrint**: **70** genes on a **DNA microarray**, high risk or low risk, **prognostic** on the evidence shown, and validated in an independent cohort of **295** patients (van de Vijver et al., *N Engl J Med* 2002;347:1999–2009).

Option by option — why each one stands or falls.

A.	fails because a hazard ratio above 1 with a confidence interval excluding 1, taken from a model of recurrence, is a statement about outcome, not about differential benefit. That analysis contains no treatment arm, so it cannot license "this patient should get this drug and that one should not". And the scale matters twice over: read as per point, the 2.81 overstates the score's power fifty-fold.
B.	fails because it is right that the recurrence figures are outcome data, and wrong to conclude from that the score cannot be predictive. The randomised B-20 arm sits in the same evidence base, and it is exactly the evidence a predictive claim requires. Reading the prognostic analysis as if it had excluded the predictive one is the trap.
C.	is right — the RS is prognostic, because it independently predicts distant recurrence (HR 2.81 per 50-point increment), and predictive, because the randomised trial with a treatment arm separated the patients who gained from chemotherapy from those who did not; the treatment arm is what licenses the second claim, and the prognostic claim does not need one. Full reasoning above.
D.	fails because it confuses validation with prediction. Validating a signature in an independent cohort — **295** patients for the **70**-gene signature, **668 of 675** archival **B-14** blocks for the **21**-gene algorithm — establishes that the score holds in a new population. It does not establish that the score identifies who benefits from chemotherapy; on the evidence shown, **MammaPrint** is prognostic only.
E.	fails because recurrence risk is prognosis. "This patient has a 30.5 % risk of distant recurrence" and "this patient gains from chemotherapy" are different claims, and only the second can justify giving a drug to some people and withholding it from others.

In systems terms. One input, two different query functions. Prognosis is a lookup — given the score, what is the baseline risk? Prediction is a differential — given the score, how much does the treatment change that risk? A differential needs two arms so that the two risks can be subtracted. The Cox hazard ratio is the lookup table; the interaction test is the subtraction. Reading one as the other is reading a lookup as a differential, and it is the most expensive mistake in biomarker work, because it licenses withholding a treatment on evidence that cannot support it.

Q3 — Answer: **B**

Why B is right. The failed parameter is the **dynamic range**. The assay's linear range is **10–100 ng/mL**; the patients' values are **100–1,000 ng/mL**. The ranges meet at 100 and do not overlap, so every value that could decide management sits at or above the top of the assay's range: for the population of interest the assay can only return "out of range" or a clipped number. The repair is to re-develop the measurement — validated specimen dilution, sample pre-treatment, or a new calibration and capture range — so that the quantification range covers 100–1,000 ng/mL, and then to re-run the analytical validation across the extended range.

And the second half is the point the course keeps returning to: **passing analytical validation licenses nothing clinically.** Analytical validation is a laboratory property — does the assay measure its target correctly and reproducibly? Clinical validity is a population property — is the measurement associated with the clinical endpoint? Clinical utility is an outcome property — does using the test improve management and outcome? Only the third changes anything for a patient.

Option by option — why each one stands or falls.

A.	fails twice. Nothing in the stem concerns cross-reactivity, and the failure mode that does — an antibody binding a homologue — is the Cadherin-17 / Cadherin-16 / Cadherin-1 case, not this one; rebuilding specificity would not make the assay read above its ceiling. And the second claim fails independently: a specific assay is still not a licensed test.
B.	is right — a dynamic-range failure, because the assay's range (10–100 ng/mL) does not cover the clinical range (100–1,000 ng/mL); the range has to be re-developed, and analytical validation alone licenses nothing clinically. Full reasoning above.
C.	fails because the LLoQ is the floor of the assay's sensitivity and this failure is at the ceiling. Lowering the LLoQ would make the assay better at detecting tiny amounts, while the problem is that the patients' values are 10 to 100 times above the top of the range. A test that cannot read the disease is not a test that needs more sensitivity at the bottom.
D.	fails because reproducibility is a real analytical parameter — the same signal across days, operators and reagent lots, which is why antibody batches made in 2024 and 2026 by a CDMO have to give the same signal. But the stem reports two ranges, not a drift, and the range failure stands however reproducible the assay is.
E.	fails because an assay whose linear range stops below the clinical range has failed, even if it behaves exactly to specification. Working to spec is not the same as being fit for purpose, and fitness for purpose is a question about the clinical range, not about the design.

In systems terms. The dynamic range is the window of the instrument. A detector that saturates cannot report a value above its ceiling; it returns a clipped, uninformative output rather than an error, which is why the clinical range has to sit inside the analytical range. The assay is a transducer with a transfer function, and any input outside its linear window is silently distorted. Then there is the ladder: rung one asks whether the instrument reads correctly, rung two whether the reading means anything in patients, rung three whether acting on it helps. A perfect instrument on rung one with nothing on rungs two and three is a device that produces numbers nobody should act on.

Q4 — Answer: **D**

Why D is right. A driver gene **causes** the oncogenic transformation; a passenger is altered, or co-expressed, and does nothing. The course licenses the driver call from the **convergence of three independent kinds of evidence** — copy-number amplification, up-regulation, and membership in an enriched pathway — because each line alone is confounded:

- **Expression alone** cannot separate them: a passenger in an active pathway is co-expressed with the driver, which is exactly the G2, G3 and G4 situation in the stem.
- **Copy-number alone** cannot either: a bystander region can be amplified by chance, with no oncogenic consequence.
- **Pathway enrichment alone** cannot either: enrichment says which biology is deregulated, not which gene inside it is causal — passengers sit in enriched pathways too.

Three agreeing lines are much harder to explain by chance than one, which is why the convergence is the call. But the call is still an **inference**, not a proof: a causal claim needs functional evidence, such as a CRISPR/Cas9 knock-out or knock-in against an **isogenic control** — a pair of cell lines genetically identical except for the single edit — read out in a human-relevant system such as patient-derived organoids.

Option by option — why each one stands or falls.

A.	fails because expression is the confounded line, not the decisive one. G2, G3 and G4 sit at similar levels, so "most highly expressed" cannot pick the driver out; the course's driver criterion has never been expression on its own.
B.	fails because amplification is one of the three lines, not the whole call, and it is confounded on its own — a bystander region can be amplified by chance — so dismissing the expression and enrichment evidence throws away the convergence that licenses the call.
C.	fails because enrichment says which biology is deregulated, not which gene in it is causal. A passenger in an enriched pathway is still a passenger, and the whole reason the course asks for expression and CNV together is that no single line decides it.
D.	is right — the driver call is licensed by the convergence of amplification, up-regulation and pathway enrichment; expression alone cannot do it, because passengers are co-expressed with drivers; and neither amplification nor enrichment is decisive on its own either. Full reasoning above.

In systems terms. The three lines are three independent sensors of one hidden state. A single sensor is confounded, two agreeing is suggestive, three agreeing is much harder to explain by chance — which is what raises the posterior. But agreement is still correlation, and correlation is not causation: only an intervention that breaks the gene (knock-out) or puts it in (knock-in), against an isogenic control, can establish the arrow. Convergence narrows the search; the experiment decides.

Q5 — Answer: **B**

Why B is right. qPCR and ddPCR answer the same question in different ways, and the ways differ exactly where this job is hardest.

- **qPCR with a standard curve** is **absolute quantification by calibration**: amplify known amounts of the target — the course's example is a two-fold series of **10, 5, 2.5 and 1.25 ng** across four tubes — and read the unknown's Ct off that curve. Lower starting template, higher Ct. The number is only as good as the calibrator and the amplification efficiency.
- **ddPCR** needs **no standard curve** at all. It partitions the sample into **thousands of droplets**, amplifies the target in each, counts the positive and negative droplets, and converts the counts into an absolute concentration by **Poisson statistics** — the correction for the fact that a droplet may carry more than one target molecule.

Three lecture reasons favour ddPCR for a rare, low-abundance, multi-target ctDNA readout: absolute quantification **without a standard curve**; **sensitivity** for the very low abundance of ctDNA; and **multiplexing** — several targets in one tube from a minimal sample, with reaction-to-reaction variation minimised, which matters when the sample is a drop of blood.

Option by option — why each one stands or falls.

A.	fails because the standard curve is what qPCR needs in order to produce an absolute number; it is not what confers rarity detection. A curve calibrated at 10, 5, 2.5 and 1.25 ng is itself dilute, and the unknown's precision inherits every error in the calibrator and in the efficiency of the reaction.
B.	is right — no standard curve, Poisson conversion of the droplet counts, and multiplexing of several targets in one small-volume reaction. Full reasoning above.
C.	fails because it inverts the methods. ddPCR is the one that gives an absolute concentration with no calibrator; qPCR is the one that needs a curve.
D.	fails because the droplet counts are converted by Poisson statistics and need no calibrator. The positive-to-negative ratio in each partition is the measurement itself, which is why the method needs no standards and why multiplexed droplets can be deconvolved.
E.	fails because the two methods do not deliver "the same answer" by construction — qPCR's absolute number depends on its standard curve and its own efficiency, ddPCR's on counting partitions — and for a rare target those are exactly the differences that matter. A method choice for a defined measurement feature is a question about mechanism, not about what the laboratory already owns.

In systems terms. qPCR is an indirect measurement: it reads a proxy (the cycle at which fluorescence crosses threshold) and reconstructs the concentration from a calibration, so the answer is a model output carrying a model error. ddPCR is a direct count: partition, count events, apply the Poisson correction for co-occupancy. For a rare target the direct count wins, because you do not have to amplify a signal into a curve in order to read it back out.

PART B — Pipeline mapping: model answers

What a full-mark answer contains, stage by stage, with the marks each limb carries and the trap each part is built around.

Q6 — Model answer (6 marks)

(a) The pipeline, stage by stage (2 marks). Name the technique, its input, its output and the stage it feeds.

1	**Pathway enrichment analysis (PEA).** Input: the list of differentially expressed genes (here 412), or for GSEA an expression-**ranked** list of all genes. Output: the enriched pathways and GO terms — biology instead of a pile of names. **g:Profiler** performs over-representation analysis on the gene list; **GSEA** asks whether an a priori gene set is concentrated at the top or bottom of the ranked list, so it does not depend on an arbitrary fold-change cut-off; **Cytoscape** builds the interaction network of the enriched pathways and their shared genes; **EnrichmentMap**, a Cytoscape app, turns the enrichment table into that overlap network, so redundant pathways cluster and one pathway feeding another becomes visible. *Cannot establish:* which gene within the pathway is causal.
2	**Copy-number analysis (CNV), read together with the enrichment result.** Input: the same tumour's copy-number profile. Output: the driver-versus-passenger call. *Cannot establish:* causation — it is an inference.
3	**Target and modality.** Input: the driver. Output: a target and a modality — a **small molecule**, which needs **target specificity**, or an **antibody**, which needs a **well-tuned affinity** (too tight and the blood half-life suffers). *Cannot establish:* that the molecule is safe, or that it can be given to a person.
4	**Hit → lead → in vitro and in vivo → preclinical toxicology → IND.** Input: the target. Output: a candidate molecule with a toxicology package: **Good Laboratory Practice** studies, including the mandatory **non-human primate (monkey)** toxicology, with the regulatory affairs specialist running the **pre-IND** meeting before the **IND** dossier goes to the **FDA** or the **NMPA**. *Cannot establish:* human safety at the clinical dose, efficacy, or the therapeutic window in patients.
5	**Phase 1 → 2 → 3 → approval, with the CDx developed alongside.** Input: the IND. Output: an approved drug with a label. Phase 1 (20–80 participants) establishes safety and dose; Phase 2 (100–300) preliminary efficacy; Phase 3 (1,000–3,000) confirmatory comparative efficacy. In parallel the **companion diagnostic** is developed and validated, because it is what selects the patients.
6	**Recruitment design and the treatment decision.** Input: a validated CDx. Output: the **enrichment design** (screen every incoming patient with the CDx and enrol only the positive ones) or the **all-comer design** (enrol everyone, measure the biomarker retrospectively), and then the **treatment decision**: the CDx result plus the label determines who receives the drug.

(b) What each stage cannot establish on its own (1 mark). Pathway enrichment cannot call a driver: it says which biology is deregulated, not which gene inside it is causal, so the CNV has to be brought in. **Copy-number analysis** cannot call one either: a bystander region can be amplified by chance with no oncogenic consequence, and up-regulation on its own may simply reflect the tumour's overall transcriptional state, because passengers are co-expressed with drivers. Only the convergence of amplification, up-regulation and pathway enrichment licenses the call — and even then it is an inference, not a proof; a causal claim needs a CRISPR experiment with an isogenic control.

(c) Between the target and the IND dossier (1 mark). The sequence is target → hit → lead → in vitro and in vivo → **preclinical toxicology → IND → Phase 1 → 2 → 3 → approval**. The toxicology has to be run under **Good Laboratory Practice** and includes the mandatory **non-human primate (monkey)** studies, and the **pre-IND meeting** comes first: it is where the sponsor submits its data and asks the regulator whether they are sufficient and what further experiments are required, before the money is spent — Dr. Wong's figure for an IND submission is about **US$1 million**, which is exactly why the meeting comes first. What the toxicology package **cannot** establish is human safety at the clinical dose, clinical efficacy, or the therapeutic window in patients. It establishes a starting dose and a risk profile in animals; establishing human safety and dose is the job of **Phase 1**.

(d) Why the CDx is developed alongside the drug, and the two recruitment designs (1 mark). The CDx is not optional and it is not a later accessory: recruitment depends on it, so it must exist, be validated and be ready **before** recruitment opens — a biomarker without a CDx is a biological observation nobody can act on, and a CDx without a validated biomarker is an instrument measuring nothing meaningful. The two designs are the **enrichment design**, which screens every incoming patient with the CDx and enrols only the biomarker-positive ones — a smaller, faster, cheaper trial that needs the CDx ready up front, but carries an initial screening burden and is wasteful for a low-prevalence biomarker — and the **all-comer design**, which enrols everyone and measures the biomarker retrospectively, which needs no CDx on day one but gives a much larger and more expensive trial.

(e) Where the chain breaks if the CDx is not ready, and what the treatment decision licenses (1 mark). The break is **between the drug and the recruitment**: with no ready CDx the enrichment design cannot open, because the screen is the entry gate, so the trial either stalls or starts as an all-comer trial and gives up the enrichment that was the economic reason for doing it at all — Dr. Wong's figure is about **US$200,000 per patient enrolled**, so enrolling patients who cannot benefit is the cost of the whole strategy. A CDx that arrives after the data are in cannot be inserted retrospectively without re-validating it on the archived samples and re-analysing. The treatment decision itself licenses: the **CDx result** plus the **label**. For the biomarker indication, the label names the biomarker, the assay and the cut-off; a label can also carry a genotype-based dosing rule, as in the FDA pharmacogenomic labelling of abrocitinib (**NDA 213871**), where **CYP2C19** poor metabolisers start at **50 mg once daily** and escalate to **100 mg once daily** if the response is inadequate after 12 weeks, because reduced metabolic clearance raises the AUC.

Q7 — Model answer (6 marks)

(a) The pipeline, stage by stage (2 marks).

1	**Sample and matrix.** Input: a blood sample, with the matrix fixed (plasma and serum are not two names for the same thing) and the pre-analytical handling recorded. Output: a sample whose upstream is known. *Cannot establish:* anything about the analyte.
2	**Extraction and quantification.** Input: the sample. Output: ctDNA quantified by **ddPCR** — thousands of droplets, positive and negative droplets counted, **Poisson statistics**, **no standard curve** — or by qPCR read against a standard curve (**10, 5, 2.5 and 1.25 ng** in the course's example). *Cannot establish:* that the number means anything clinically.
3	**Analytical validation.** Input: the assay plus its validation experiments. Output: a documented analytical performance — **sensitivity** down to the **lower limit of quantification (LLoQ)**, **specificity** and cross-reactivity, **dynamic range**, **reproducibility** across days, operators and reagent lots, and **matrix effect**. *Cannot establish:* clinical validity or clinical utility.
4	**Clinical validation.** Input: the validated assay plus a labelled patient cohort. Output: an **ROC curve**, an AUC, and a chosen cut-off with its sensitivity and specificity. The miRNA classifier is the course's worked example: **miR-15b + miR-130b**, discovery in **96** HCC tumours with matched adjacent non-tumour tissue, a **15**-patient longitudinal subset in which serum levels fell after the tumour was removed, and an independent validation cohort of **30** healthy controls, **29** chronic hepatitis B carriers and **57** HCC patients, giving **AUC 0.981**, **sensitivity 98.25 %**, **specificity 91.53 %** and **accuracy 94.83 %**. *Cannot establish:* clinical utility.
5	**Clinical utility.** Input: a randomised comparison of management with the test against management without it. Output: evidence that acting on the test improves outcomes.
6	**Regulatory label.** Input: the utility evidence. Output: the label — the biomarker, the assay, the cut-off, the population and the regimen.

(b) The analytical parameters, what each catches, and what the package cannot establish (1 mark).

Sensitivity and the LLoQ catch a floor problem: the lowest amount the assay can reliably quantify. **Specificity and cross-reactivity** catch a false-signal problem: for protein immunoassays homology between family members is the trap — an antibody raised against Cadherin-17 must be counter-screened against Cadherin-16 and Cadherin-1 — whereas a nucleic-acid assay gets its specificity from base pairing and probe design. **Dynamic range** catches the ceiling problem: an assay reading 10–100 ng/mL is useless when patients carry 100–1,000 ng/mL. **Reproducibility** catches drift: the same signal across different days, different operators and different reagent lots, which is why antibody batches made in 2024 and 2026 by a CDMO have to give the same signal. **Matrix effect** catches interference from the specimen itself — intrinsic (the specimen's own albumin and immunoglobulins) and extrinsic (container and reagents) — with the two special cases the course names: an **autoantibody** to a tumour neoantigen **amplifies** the signal and makes a good biomarker, whereas an **anti-drug antibody (ADA) interferes**, competing with the assay's capture and detection antibodies and able to neutralise the drug. What the whole package still cannot establish is clinical validity and clinical utility: it is a laboratory property, not a population property.

(c) Clinical validity, how it is established, and what it does not license (1 mark). Clinical validity is the demonstration that the measurement is **associated with the clinical endpoint**, established by an **ROC curve** over a labelled cohort, with a cut-off chosen and the sensitivity and specificity at that cut-off reported. The worked example shows it: the **miR-15b + miR-130b** classifier of Liu AM et al. (*BMJ Open* 2012;2:e000825) gave **AUC 0.981** in an independent cohort of **30** healthy controls, **29** chronic hepatitis B carriers and **57** HCC patients, with **sensitivity 98.25 %**, **specificity 91.53 %** and **accuracy 94.83 %**; the **96**-patient discovery step and the **15**-patient longitudinal subset (levels falling after tumour removal) supply the evidence that the signal is tumour-derived rather than a constant of the assay. What clinical validity still does not license is a treatment decision: it says the measurement correlates with the endpoint, not that using the test improves management or outcome. That is clinical utility, and it is the only rung that changes anything for a patient.

(d) Clinical utility, and why an analytically perfect assay with no utility changes nothing (1 mark). Clinical utility is the demonstration that **using the test improves patient management and outcome**, established by a trial comparing management with the test against management without it. An analytically perfect assay with no clinical utility changes nothing for a patient because the assay is only an instrument: a measurement becomes medicine when someone can act on it and the acting improves the outcome. Without utility evidence the test may be accurate and associated with the endpoint and still be useless, because nobody is better off for holding the number — the course's own sentence is the exam answer: a biomarker is useless unless it has clinical utility. The ladder has to be climbed in order: analytical validation proves the assay measures its target, clinical validity proves it correlates with the endpoint, and only clinical utility proves that acting on it helps.

(e) The label, and the joint FDA–NIH resource (1 mark). The label has to say **which biomarker, which assay, which cut-off, which population and which regimen** — the assay is named because it is the CDx that defines the trial population, and the cut-off is named because it is part of the claim; where the biomarker also carries a genotype-based dosing rule, the label carries that too (the abrocitinib example: **NDA 213871**, where **CYP2C19** poor metabolisers start at **50 mg once daily** and escalate to **100 mg once daily** after 12 weeks if the response is inadequate, because reduced metabolic clearance raises the AUC). The joint FDA–NIH resource used to qualify biomarkers and endpoints is **BEST** — **Biomarkers, EndpointS, and other Tools** — the joint resource of the **FDA** and the **NIH** Biomarker Working Group (NCBI Bookshelf NBK326791). It is not an NCI resource, and it is what a sponsor uses to state the context of use and the evidence a biomarker needs in order to be qualified for its intended purpose.

PART C — Exception handling: model answer

Where the system breaks, what that does to every observation in the table, and how you would prove where the break is.

Q8 — Model answer (8 marks)

(a) Locate the break and trace the consequence (3 marks).

The pipeline for this programme is **biology → biomarker → CDx → analytical validation → clinical validity → clinical utility → trial → label → treatment decision**, and the break is **not** in the assay.

- **What has passed: the CDx.** Every analytical parameter holds — lower limit of quantification, specificity and cross-reactivity, dynamic range, reproducibility across days, operators and reagent lots, and matrix effect including autoantibody and anti-drug antibody interference. The assay measures what it claims to measure, reproducibly, in the matrix it will be run in, and it was applied prospectively at screening with the cut-off locked before randomisation. There is no measurement problem to find here, which is why hunting for one wastes the week.
- **What has failed: the rung between clinical validity and clinical utility.** The biomarker does not identify the patients who benefit — it identifies the patients who do **not** benefit from this antibody. The interaction test is significant (P = 0.02), but it runs the other way: biomarker-negative patients gain (HR 0.58, 95 % CI 0.38–0.88, P = 0.008), while biomarker-positive patients gain nothing over chemotherapy alone (HR 1.06, 95 % CI 0.80–1.41, P = 0.69). The sponsor's hypothesis is not merely unproven; it is inverted.
- **The downstream consequences.**
- *The CDx*: it cannot be used as a selection criterion for this drug. Either the biomarker is redefined, or the assay is repurposed (as a prognostic or monitoring readout), and the protocol has to say which.
- *The trial*: the proposed confirmatory trial must not enrol biomarker-positive patients only. That design would take the population in which the antibody has already shown no benefit over chemotherapy and enrol it at full cost — the worst possible use of a trial.
- *The label*: no biomarker-restricted indication can be filed on this biomarker. Any indication has to rest on the all-comer evidence, and the CDx is not a selection device for it.
- *The biomarker hypothesis*: it has to be rewritten. Either the biology the CDx detects is a resistance mechanism — in which case the finding is a fact about the drug, not a defect of the assay — or the direction of the claim was mis-specified

in the protocol, or the cut-off and the assay's classification in this population have to be audited before anything is concluded.

(b) What the biomarker was actually shown to predict (1 mark). It was shown to be **predictive** — of the opposite thing to the hypothesis. Biomarker-**negative** patients gain from this antibody (HR 0.58, 95 % CI 0.38–0.88) while biomarker-**positive** patients gain nothing over chemotherapy alone (HR 1.06, 95 % CI 0.80–1.41), and the biomarker-by-treatment interaction is significant (P = 0.02). So the biomarker predicts differential benefit from this drug, but in the reverse direction, and used as a selection rule it would exclude the responders. What it was **not** shown to do: it was not shown to identify who benefits — that is the claim the sponsor made and this trial refuted; it was not shown to be merely prognostic; and it was not shown that biomarker-positive patients are harmed, because no benefit over chemotherapy is not the same as harm.

(c) What evidence a predictive claim needs, and from which rung (2 marks). A predictive claim needs a **treatment arm** — differential benefit cannot be demonstrated without a randomised comparison — and it needs the claim itself to be pre-specified, with the assay applied prospectively at a **locked cut-off** in the population the claim is made about.

1	**A prospective, randomised, adequately powered trial in the population the claim is made for.** For the sponsor's original claim that is impossible from this trial, because this trial has already refuted it. For the reversed claim it is a new trial in biomarker-negative patients, with the CDx run centrally at screening, the cut-off locked, the interaction test formally significant, and the subgroup pre-specified rather than post hoc.
2	**Utility, not merely validity.** The label has to show that **using** the test changes management and improves outcome, not only that the measurement correlates with an endpoint. This is the rung the current evidence is missing.
3	**An audit before any of that.** The assay's real performance in the trial population (LLoQ, call rate, concordance with any local testing), and the biological question of whether the biomarker detected is a resistance mechanism — because if it is, the "predictive" finding is a fact about the drug and not a useful selection rule at all.

The rung to climb is therefore **clinical utility**, on top of a clinical validity that has already been established — in the reversed direction.

(d) What an enrichment design controls for, and what it cannot (1 mark). An enrichment design **controls for** assay readiness and uniformity — every patient classified by the same validated assay at the same cut-off before randomisation — and for the **size, speed and cost** of the trial: it enrols only patients expected to respond, which is the economics Dr. Wong used (**US$200,000 per patient enrolled**). It **cannot control for** whether the biomarker is predictive at all, whether the cut-off is the right cut-off, or whether the hypothesis is right, and it cannot substitute for a treatment arm, because a trial with no comparison arm has no interaction to test. Had the sponsor used one here, the trial would have enrolled biomarker-positive patients only — precisely the group in which the antibody showed no benefit over chemotherapy (HR 1.06, P = 0.69) — and it would have produced a clean, well-powered, entirely uninformative null, having spent the cost of every patient on the population the biomarker had wrongly identified as the responders. That is the design's worst case, and it is invisible until the data come in.

(e) What the team should do next, in order (1 mark). In order: (1) **stop** — do not file a biomarker-restricted label and do not launch the enrichment trial; (2) **audit the measurement** — assay performance and call rates in the trial population, the locked cut-off, concordance between the central lab and any local testing, and the pre-specification of both subgroup analyses; (3) **run the formal interaction analysis** and ask whether the biomarker is a resistance mechanism rather than a response marker, and whether the prognostic hazard across both arms runs the other way; (4) **decide the biomarker hypothesis** — redefine it, reverse the direction of the claim, or drop it; (5) **re-plan the programme** — if the reversed claim survives the audit, run a prospective, pre-specified randomised trial in biomarker-negative patients and qualify the biomarker against the **FDA–NIH BEST** framework; (6) **tell the regulator early**, because an interaction that reverses a selection biomarker changes the risk–benefit of the whole programme, including for the patients already enrolled.

Self-marking scale

Band	What it looks like
25–30	You are answering from mechanism, not recall. Chains are correct and the limits are named.
20–24	Solid on Parts A and B; the weak spot is usually Part C — you found the fault but stopped before following it downstream.
15–19	You know the content but not the wiring. Re-do the two pipeline questions and draw the chain before writing.
Below 15	Work the Lecture 7 chains (the validation ladder, driver vs passenger, prognostic vs predictive, qPCR vs ddPCR) and the case bank, then sit this paper again.

What happens next

Submit your answers and the grading is done properly: a full post-mortem, question by question, with every biological error translated into computational or data-flow terms — what the mistake would look like if the mechanism were code, and which part of the pipeline your reasoning broke at.

Source basis: `GNBF5020_Complete_Revision_All_Lectures.md` *(Lectures 1–5, audio + slides),* `GNBF5020_Lecture_6_Exam_Focus_and_Learning_Guide.md` *(Lecture 6),* `GNBF5020_Lecture_7_Exam_Focus_and_Learning_Guide.md` *(Lecture 7, including its Appendix C corrections),* `GNBF5020_Case_Studies.md`*. The microarray-vs-RNA-seq cover plate in the Lecture 6 guide is sourced to Rao MS et al., Frontiers in Genetics 2019;9:636 (doi:10.3389/fgene.2018.00636). The question paper is* `GNBF5020_Practice_Exam_02_Cumulative_L1–L7.pdf`*.*